

The usability of a novel mobile augmented reality application for excavation process considering safety and productivity in construction

Mobile
augmented
reality
application

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Fathima Nishara Abdeen

*School of Built Environment, Faculty of Arts, Design and Architecture,
University of New South Wales, Sydney, Australia*

Randima Nirmal Gunatilaka
CBRE Pty Limited, Sydney, Australia

Samad M.E. Sepasgozar
*School of Built Environment, Faculty of Arts, Design and Architecture,
University of New South Wales, Sydney, Australia, and*

David John Edwards
*Department of the Built Environment, Birmingham City University,
Birmingham, UK and Faculty of Engineering and the Built Environment,
University of Johannesburg, Johannesburg, South Africa*

Abstract

Purpose – This study aims to assess the usability of augmented reality (AR) based mobile app for excavation and earthmoving processes using a novel tool entitled Excavator Augmented Reality (EAR).

Design/methodology/approach – A mixed-methods research approach was used through conducting experimentation to collect qualitative and quantitative data collected from the Sri Lankan construction sector. EAR app was used for experimentation in outdoor areas examining how a 360° tracked hydraulic excavator can be navigated in different physical environments similar to the real prospected job.

Findings – The findings reveal that EAR could make a considerable impact on enhancing productivity, safety and training processes. However, the developed EAR App subjected to assessment demonstrated the highest satisfaction gap for the auditory aspects. Among the remaining criterion, the satisfaction met user expectations for comfortability and no-risk practice. An analysis of strengths, weaknesses, opportunities and threats (SWOT analysis) conducted revealed that visualising the excavator activities and the requirements of improved features were the highest agreed strengths and weaknesses of the EAR. Among the opportunities for improvement, the necessity of improving emergency and safety reached the highest agreement. Moreover, the study presented the challenges in introducing mobile augmented reality (MAR) to the construction sector under the political, economic, sociocultural, technological, environmental and legal (PESTEL) model along with solutions to be taken.

Originality/value – This study provides a novel approach to addressing the safety, productivity and training concerns in heavy mobile plants and machinery on construction sites which remains to be unexplored to this end.

Keywords Mobile augmented reality, Augmented reality, Excavator, Safety, Construction

Paper type Research paper



1. Introduction

The literature shows that the frequency and severity of accidents occurring on construction sites remain unacceptably high, mainly due to the inherent hazards and dangers associated with the works undertaken (Singh and Misra, 2021). Moreover, the industry is highly labour intensive, and activities are difficult to supervise given the continuously changing nature of construction sites and the multitude of site dangers (Ng and Tang, 2010). These complexities make the construction sector one of the most dangerous globally, with fatalities and accidents occurring regularly. Against this contextual backdrop, information visualisation techniques such as building information modelling (BIM), virtual reality (VR) augmented reality (AR), as well as other game engine-based mixed reality (MR) techniques, have been proposed to advance the current safety management practices (Gunatilaka *et al.*, 2021; Li *et al.*, 2018). AR applications, in particular, are both ubiquitous and important in the digitisation of construction processes, especially in the paradigms of safety (Lovreglio and Kinateder, 2020) educational, architectural and field engineering processes for both students and professionals alike (Behzadi, 2016). Moreover, AR in construction could also minimise the cost incurred due to poor time management and perform iterative processes while providing a variety of professions with an easy, cost-effective and safe system (Behzadi, 2016), implying its contribution to safety and productivity. Further, a noteworthy increase in mobile augmented reality (MAR) applications is evident in the construction sector. The construction industry relies on several different construction trades, such as masons, carpenters, plumbers, steel benders and painters (Ekow and Kofi, 2016). Each of them has work items that cross over periodically during the construction of a project – for example, carpenters and bricklayers could work together in foundation work to build shuttering and lay bricks. In such a scenario, human-induced errors will significantly affect the efficiency of the processes. Technologically facilitated construction related apps would facilitate in enhancing the jobsite safety and productivity and minimise the challenges faced by the members of project teams (Azhar and Cox, 2015).

Experts predict the investment AR/VR industry to reach more than \$25bn by 2025 (Makarov, 2020). However, only a small number of AR and VR studies (under 15%) used handheld and mobile devices such as smartphones as their “Computing Units” or “Enabling Technologies” for their research (Piroozfar *et al.*, 2018). Moreover, most apps used in the construction industry are restricted to data or document viewing (Azhar *et al.*, 2015). Nevertheless, there is a growing requirement to use AR technologies in heavy mobile plants and machinery (Sitompul and Wallmyr, 2019) such as rough terrain telescopic handlers and articulated dump trucks (Edwards *et al.*, 2021). However, augmentation in the excavation has been concentrated more towards head-up displays (Wallmyr *et al.*, 2018), video-based displays (Akyeampong *et al.*, 2012) and projection displays (Behzadan and Kamat, 2009), overseen the use of MAR application to enhance the productivity, efficiency and safety of excavation process. Moreover, within the Sri Lankan context, there is a dearth of MAR studies applied to excavation processes. Consequently, this research aims to assess the usability of an AR-based Mobile App for the excavation process called Excavator Augmented Reality mobile app (EAR) and address the following research questions:

- RQ1. What are the expectations and current level of satisfaction concerning the developed EAR?
- RQ2. What are the implications of the EAR App for the construction industry?
- RQ3. What are the strengths, weaknesses and opportunities for improvement of the EAR App?

RQ4. What are the wider political, economic, social, technological, environmental and legal challenges associated with such applications and solutions?

2. Literature review

2.1 Mobile augmented reality applications in construction industry

Despite significant efforts and careful consideration from governmental organisations and professional bodies, the current safety practices rely on parametric models or manual methods that involve some infrequent, ineffective and error-prone auditing and poor safety training (Park *et al.*, 2020). These practices may enhance safety, but they negatively affect the productivity of the project. However, developing cutting-edge technologies may improve the construction industry's safety, efficiency and productivity. Some examples of digital technology applications include sensors for alarm and awareness that may prevent harmful exposures; radar sensors and AR to control heavy machinery onsite; drones to perform site surveys; mixed virtual and AR for training; and the use of smart wearable devices (smart watches, vests) to track employees (Cartwright, 2018). Further, the introduction of three-dimensional (3D) printing and robotics has taken a drastic advancement with the capability of creating building components or the entire building. Hence, the last 10 years have seen the introduction of various tech-based alternatives to control and lower site risks and enhance productivity (Park *et al.*, 2020). These solutions include BIM, VR/AR, artificial intelligence (AI), computer vision and big data analytics.

However, in particular, there is a trend driving technological advancement directed at modelling AR based on mobile devices, which enables 3D models to be configured and placed in environments without markers being required (Machado and Vilela, 2020). AR is proving to be useful in many other sectors, and with the advent of smaller and more efficient mobile electronic devices, AR can also become a real tool for construction professionals (Kim and Irizarry, 2020). MAR applications in the construction industry enable a wide range of stakeholders fast and easy access to information leading towards corrective actions, which minimise cost and delays to performance incompatibilities (Bae *et al.*, 2013).

Veas *et al.* (2013) developed a platform for 3D mobile viewing of environmental data integrated with AR and used for monitoring the construction environment. Yeh *et al.* (2012) in their study used mixed reality as a remedy to minimise the complexities and difficulties of onsite data retrieval in construction projects. Further, the application of mixed reality apps in detecting dimension errors and omissions is discussed in a study conducted by Kwon *et al.* (2014). Further, Kim and Irizarry (2020) also used MAR as an innovative tool to enhance the spatial skills of construction management students. They tested a MAR application based on a sample of students, and feedback was obtained against a set of criteria mental demand; physical demand; temporal demand; performance; effort; and frustration. Another potential use case of MAR is its applicability in on-site construction inspection. To facilitate this requirement Kopsida and Brilakis (2016) proposed a framework based on mobile-based AR which supplements the view of the inspector with the 3D as planned BIM model.

In a different project, Chu *et al.* (2018) used a BIM integrated AR system for mobile devices to enable onsite information extraction to reduce the propensity of workers to commit cognitive failure; this work was, however, limited to an artificial scenario. Similarly, by integrating BIM into AR, an AR mobile channel called "BIM-Phase" was proposed by Zaher *et al.* (2018) for monitoring the progress of construction work. Kim *et al.* (2016) propose a system for enhancing construction safety education through the use of mobile-based VR and AR for experiential learning.

2.2 Mobile augmented reality in excavation process

Since the release of the first sophisticated wearable VR/AR device (Google Glass, Forte VFX1), the realism of mobile MR devices has seemed imminent, with the ability to enhance the way in which information is accessible and conveyed (Wang *et al.*, 2018). Recent advances in MR technologies offer new solutions to the current needs of construction projects with the aim to enhance safety and productivity by training labours to learn how to work with equipment and identify hazards in various construction conditions. As a result, the research looked into the present state of the art in terms of MAR and heavy machinery integrations.

Previous studies reveal that MAR has been applied in construction for various purposes such as onsite data retrieval (Yeh *et al.*, 2012); defect management (Kwon *et al.*, 2014); learning (Kim and Irizarry, 2020); and progress monitoring (Chu *et al.*, 2018). Although these publications demonstrated the usefulness of MR applications, the implementation evaluation of the online MR applications in the construction context is not fully explored. In particular, the application of MAR for the excavator operation for underground activities or excavation processes has not been fully investigated, which potentially can enhance both measures of safety and productivity. The study by Talmaki *et al.* (2010) developed a framework for underground utility locating methods during the excavation process by using AR. However, the paper mostly focuses on the registration process of superimposing pipeline models on the real scene using a camera reference system. The use of AR visualisation and the Global Positioning System (GPS) to generate real-time views of an excavation site is investigated by the study of Behzadan and Kamat (2009). In the study, computer-aided design models of underground utilities are accurately overlaid over live video streams of the real world, with the resulting views being displayed to the equipment operator in real time. The augmented scene has been created through a head mounted display in the study. However, it would be a worthwhile effort to look at the possibility of integrating the advancing mobile devices to recreate augmented mobile reality. Hansen (2021) showed how 3D reconstructions (usually coloured point clouds) of excavations and global information system (GIS) data can be integrated into onsite visualisation with an AR system as a prerequisite for planning or on-surface marking. Many studies look into augmenting underground utilities as construction managers need to rely on GIS data obtained from relevant authorities to avoid damages to underground utilities during the excavation process. However, the utilisation of MAR to augment excavator operations has not grasped considerable academic attention. Yet, it is important to examine how MAR improves machine operation by superimposing computer-generated content while not intervening with the mobility of an operator to perform a range of operational tasks (Abbas *et al.*, 2020). Therefore, to fill the existing gap, the current study focuses on assessing such applications through a usability study of a novel MAR application for the excavation process in construction.

3. Methodology

In addressing the research questions of the study, both quantitative and qualitative data are generated. The mixed methods approach best suits studies where quantitative and qualitative data are generated (Taylor *et al.*, 2015). Further, this method support researcher in better comprehending the connections or discrepancies between qualitative and quantitative data (Shorten and Smith, 2017). Moreover, qualitative data can add more detailed contextualised insights while enriching the quantitative results. Hence, the mixed methods approach was used to acquire complementary information on how the AR

application is used and to seek new perspectives or in-depth information about the quantitative outcomes.

Figure 1 shows different steps of the investigations, including using the novel EAR app developed and examined by a range of participants from Sri Lanka.

3.1 Interview structure and design

There is a lack of evaluation methods relevant to AR mobile apps. Hence, quantitative data metrics developed by van Baren and Ijsselstein (2004) on measurement approaches were used. This provided the researchers with an avenue to comprehend the earlier approaches to understanding users' expectations and satisfaction and adapt the tool to meet the study's purpose. A 5-point Likert scale was used to codify the answers, which despite its drawbacks, is a very successful way of codifying the response (Willits *et al.*, 2016). Qualitative open-ended questions were also incorporated to understand the reasons behind the evaluation responses and explore the potential uses, challenges of adoption, strengths, weaknesses of the EAR app and opportunities for improvements. The semi-structured interview questions are as follows (Table 1).

3.2 Data collection and dissemination

Under convenience sampling, 32 experts were selected who have an understanding of AR and the excavation process. These experts were from two large construction projects; two small and medium construction projects; and one international vocational and technical training institute. A purposive convenience sampling technique was used as it enabled the selection of experts from the defined categories and informative to answer the questions.

The respondent profile is presented in Table 2.

The experts were initially presented with the EAR app and a tutorial supporting them to operate the app (Figure 2).

Subsequently, respondents were required to rate the EAR app and provide answers to qualitative questions. The interviews were carried out face-to-face and in-person for a period ranging between 45 min to 1 h.

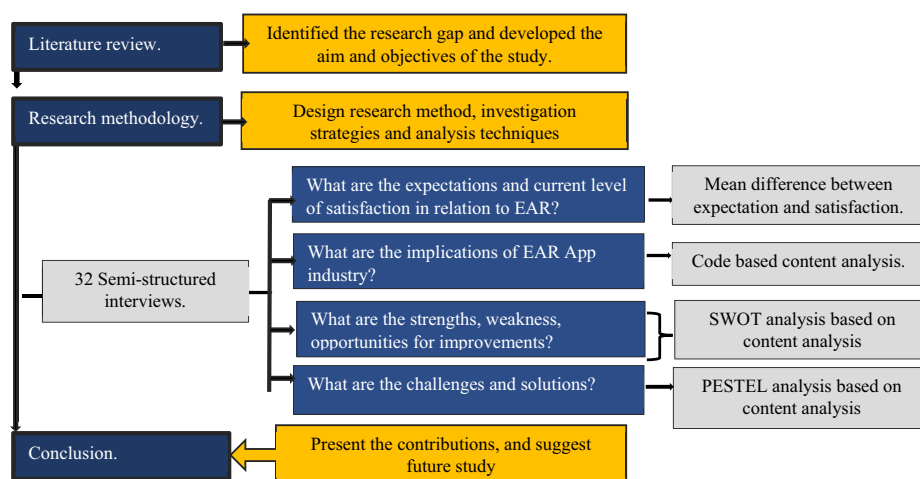


Figure 1.
Research
methodology,
including the field
experimentations and
interviews

Table 1.
List of factors suggested by the literature for examining the participants’ experience using the ERA

<i>Please mark with an “X” to rate the following statements on a scale of 1–5 (expectation: 1 = not at all important, 2 = slightly important, 3 = neutral, 4 = important, 5 = utmost important/satisfaction: 1 = completely dissatisfied, 2 = dissatisfied, 3 = neutral, 4 = satisfied and 5 = completely satisfied)</i>		
Relevant factors suggested by the literature	Sources	Statements
Enjoyment	(Carrera <i>et al.</i> , 2018); (Choi, 2019)	I completely enjoyed the virtual world
Usability	(Carrera <i>et al.</i> , 2018); (Choi, 2019); (Ohlei <i>et al.</i> , 2018)	I felt very comfortable when using the EAR App
Skill development	(Carrera <i>et al.</i> , 2018); (Choi, 2019); (Ronyastra, Hapsari and Pani, 2019)	It was useful in developing the skill
Value/usefulness/ease of learn		It was useful for learning the construction equipment movement capability
Risk involved	(Gerhard <i>et al.</i> , 2001); (Ohlei <i>et al.</i> , 2018)	It gives a no-risk practice
Motivation	(Carrera <i>et al.</i> , 2018); (Ronyastra, Hapsari and Pani, 2019); (Ohlei <i>et al.</i> , 2018)	I would like to repeat the virtual experience
Immersion	(Geszten <i>et al.</i> , 2018); (Silvennoinen, Vogel and Kujala, 2014)	I had a sense of acting in the virtual space, rather than operating something from outside
Visual involvement		The visual aspects of the environment completely got me involved
Auditory involvement	(Han <i>et al.</i> , 2021); (Kim and Jin, 2015)	The auditory aspects of the environment completely got me involved
Consistency with reality	(Han <i>et al.</i> , 2021); (Geszten <i>et al.</i> , 2018)	My experiences in the virtual environment seem consistent with real-world experiences
Controllability of actions	(Han <i>et al.</i> , 2021); (Ohlei <i>et al.</i> , 2018)	I was able to control the events in EAR
<i>Please answer the following questions</i>		
Relevant factors suggested by the literature	Sources	Questions
Learning experience	(Videnovik <i>et al.</i> , 2020); (Pal and Vanijja, 2020)	How EAR improved your learning experience? What was useful?
Involvement and immersion	(Gerhard <i>et al.</i> , 2001); (Pal and Vanijja, 2020)	Have you become involved and immersed in excavator running and been able to focus on one specific activity such as running or moving the arm?
Productivity	(Youngblut and Perrin, 2002); (Pal and Vanijja, 2020)	Can EAR help to improve the productivity of the excavation process, safety operation and training processes? If so, how?
Safety		
Training process	(Gerhard <i>et al.</i> , 2001); (Dirin, 2020)	What are the strengths and weaknesses of the EAR App?
Attitudes, beliefs and experiences		What are the opportunities for improvement of the EAR App? Give examples of what can be added as a new feature?
		Can you give an example of each political, economic, social, technological, and legal challenge in using EAR App in particular?
		What are the suggestions to overcome the challenges?

Respondents	Position	Experience
R1	Project manager	12
R2	Project manager	15
R3	Site manager	08
R4	Site manager	11
R5	Equipment manager	14
R6	Equipment manager	16
R7	Equipment manager	15
R8	Health and Safety Officer	10
R9	Health and Safety Officer	10
R10	Health and Safety Officer	09
R11	Health and Safety Officer	12
R12	Health and Safety Officer	16
R13	Chief engineer	21
R14	Chief engineer	17
R15	Excavator operator's trainers	08
R16	Excavator operator's trainers	11
R17	Excavator operator's trainers	12
R18	Excavator operator	07
R19	Excavator operator	12
R20	Excavator operator	11
R21	Excavator operator	14
R22	Excavator operator	14
R23	Excavator operator	13
R24	Excavator operator	18
R25	Excavator operator	20
R26	Excavator operator	22
R27	Excavator operator	19
R28	Excavator operator	14
R29	Excavator operator	16
R30	Excavator operator	08
R31	Excavator operator	10
R32	Excavator operator	11

Notes: Each respondent is referred to by R throughout the paper, e.g. R1 refers to the first participants that have been interviewed

Table 2.
Respondent profile

3.3 Data analysis

The analysis of quantitative data was carried out using means to understand the satisfaction gap on the developed EAR app. The analysis of qualitative data was performed through content analysis, strengths, weaknesses, opportunities and threats (SWOT analysis) and political, economic, sociocultural, technological, environmental and legal (PESTEL) analysis. The implications of the EAR app application were analysed through content analysis. Content analysis was chosen as it is one of the desired methods to analyse qualitative data (White and Marsh, 2006). The process was performed manually in which initially, text in transcripts was divided into meaning units and condensed further while keeping the central meaning; codes were formed by forming labels to describe the condensing unit; categories are then formed to group codes similar in content and context; and themes are formed if latent content is visible in two or more categories.

The SWOT analysis has been widely used in different fields to analyse construction practices, technologies and innovations from a scientific viewpoint (Nikjow *et al.*, 2021). According to the authors, both internal and external variables are included in a SWOT

study in which opportunities and thetas are captured in external factors. However, when constructing a SWOT diagram, a detailed review of external variables is not carried out (Nishadha, 2021). At least not to the extent that a PESTEL analysis would (Nishadha, 2021). As a result, to understand the threats prevailing in EAR App adoption, a PESTEL analysis was carried out (Teisserenc and Sepasgozar, 2021). SWOT and PESTEL analysis were performed using content analysis. Codes were formed based on the condensed unit and categories were developed in reference to SWOT and PESTEL analysis. As an example, the condensed meaning “accurate controls” was considered as a code under the category of strength.

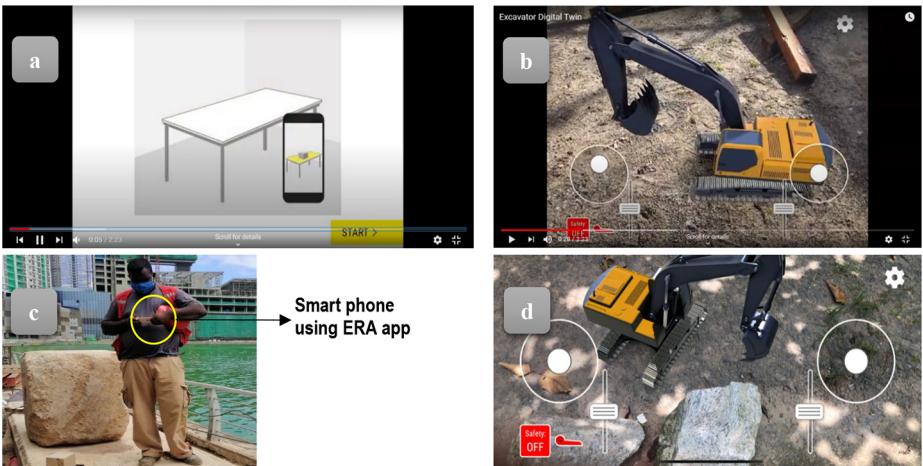
4. Research findings and discussions

4.1 Excavator augmented reality app – satisfaction gap

The satisfaction gap computed based on the differences in mean of satisfaction and expectation of the EAR app is presented in Figure 3. The results manifested that the highest value for the satisfaction gap (3.4) was demonstrated for the auditory aspects, in which 94% ($f = 30$) of the respondents rated completely dissatisfied or dissatisfied. In this respect, several respondents provided their views such as “I can’t hear anything”, “are sounds working”, “why haven’t you included sounds”, “sounds are essential” and “sounds provide more reality”. A complete discussion regarding this was provided while discussing strengths, weaknesses and opportunities for improvement in the EAR. The second highest satisfaction gap (0.9) was recorded for the usefulness of learning construction equipment capability. Though the gap was not considerable, around 22% ($f = 7$) of participants who rated dissatisfied and 41% ($f = 13$) who were neutral argue that the app should be further improved to get the right potential. These are discussed in detail in section 4.3. The satisfaction gap for the visual aspects was ranked third with 0.8125 and the 4th, 5th, 6th, 7th, 8th, 9th, 10th, 11th ranks did not demonstrate a major satisfaction gap; in fact, the satisfaction for the 11th ranked criteria, which is on risk practice exceeded the expectations of the respondents.

Figure 2.

Experimentation in a selected construction site: (a) the app environment for searching the ground to start the AR experiment; (b) showing previous experimentation to the participants, where conducted by the investigators and uploaded on YouTube; (c) a participant used EAR before the interview; and (d) demonstration of the participant effort running the excavator in a congested area



Though the respondents were satisfied in general with EAR, the existing gaps represent that several improvements should be incepted to reach the industry's expectations. These improvements suggested by respondents are subjected to discussion in the following sections.

4.2 Implications of excavator augmented reality application in construction

Respondents believe that the EAR App could deliver several positive attributes that could benefit the construction industry. However, most of the respondents (78%) perceived that the application within the current status would only have limited benefits, but an improved version would bring several impacts to the construction industry. As an example:

Sri Lanka is way far than other developed countries in terms of such technological advancement. So these initiatives would add value in the long run. (R4)

The rest (7/32) had contradictory views on the extent to which EAR might be useful:

I don't see why we should focus on such apps. In the current form of construction sites with all the types of complexities, there are much more we should consider. (R4)

These opposing views, while not denying the importance of EAR provide some evidence of the resistance to change prevailing among practitioners. However, the contribution of the App to the construction industry at large was recognised by the respondents. By performing content analysis the study identified three major themes around which the implication of EAR application to construction is based on. These themes, which are safety, productivity and training, are presented in Figure 4 and explained below.

4.2.1 Safety. The main implications of the app under ensuing safety are minimise accidents; assist in the inspection; hazard management; and familiarise emergency procedures.

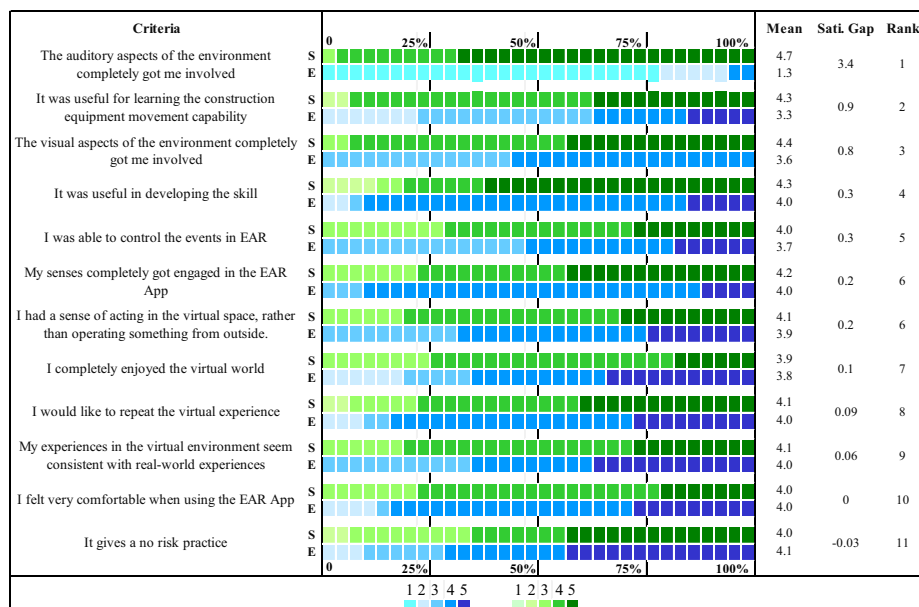


Figure 3. Satisfaction gap, (scales: expectation – 1 [not at all important], 2 [slightly important], 3 [neutral], 4 [important], 5 [utmost important]/satisfaction-1 [completely dissatisfied], 2 [dissatisfied], 3 [neutral], 4 [satisfied], 5 [completely satisfied])

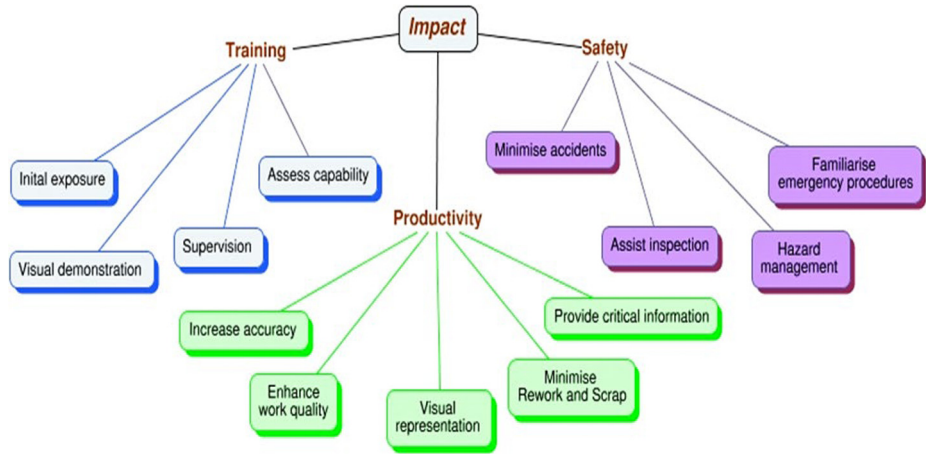


Figure 4.
Implications for
construction industry

Respondents strongly believe that digital representation through the application will aid in ensuring safety in construction sites. As an example, health and safety officers (R8-R12) strongly believe that the EAR app could contribute to safety aspects by minimising accidents during excavator training and familiarising with emergency procedures and inspections before the operation. Respondent R8 pointed out “the App provides a basic platform for training, compared to onsite training where the possible hazards are comparatively low”. Moreover, the possibility of using this application to familiarise with emergency procedures and inspections mandated by all the health and safety officers and trainers. The approach of MAR was indicated by the excavator operators (R20, R25–R30) as an opportunity for hazard management through awareness generation by creating hazard simulations.

4.2.2 Productivity. Themes relating to productivity appeared within the analysis, with several participants talking about the impact on accuracy, work quality, visual representation, minimising rework and providing critical information. The increased productivity is an indication of the successful application of the app in streamlining real-life challenges. Project managers (R1–R3) and engineers (R13–R14), in particular, noted that a well-developed EAR app could provide a detailed understanding of the excavator functionality visually. Moreover, it was stated that it could give workers critical information as they perform operations and repairs while improving accuracy and work quality and reducing rework and scrap.

4.2.3 Training. As presented in Figure 4, training is considered a major contribution of the EAR app to the construction industry. It was agreed that the app could facilitate obtaining an initial experience in excavator operation and controls, provide a platform to visually demonstrate the task without visiting the site at the initial stages, and supervisors can mentor and assess an operator’s capability and familiarity with controls. While the app can be subjected to further developments, it could also be used to enhance the participants’ training experience. In this regard, excavator operators and trainers noted that the app, in its current status, is insufficient for training purposes, yet if developed further, it would very well cope with the training requirements.

4.3 Strength, weakness and opportunities for improvements

Several considerations arose during interviews regarding EAR strengths, weaknesses and opportunities for improvement. These findings are summarised in Figure 5.

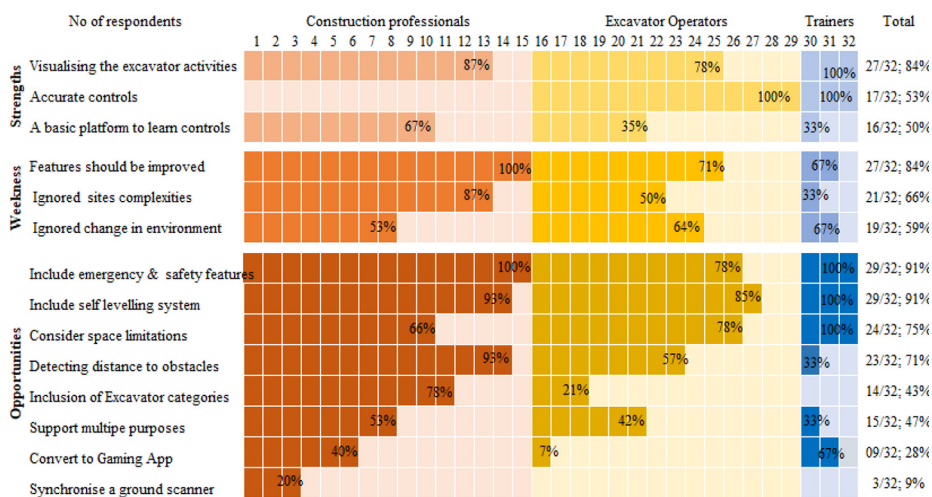


Figure 5.
Strength, weakness
and opportunities for
improvements

Figure 5 represents the main categories derived based on the codes developed through the condensed meaning units extracted through interviews. The darker coloured boxes in brown, yellow and blue indicate the frequency of a code. For instance, 13 out of 15 construction professionals have agreed on the capability of the EAR app on “visualisation exaction activities”, which was categorised as a strength. Each column represents the types of experts considered in the study. The total column provides the aggregated value, which considers the level of agreement by all the experts.

4.3.1 Strengths. The majority of respondents (84% or $f = 27$) strongly believed the ability of the EAR app to visualise the excavator activities was one of the key strengths. The next most agreed (53% or $f = 17$) strength of the EAR App by all excavator operators and trainers is its accurate controls. For example, R21 stated:

[. . .] all controls, including arms and bucket, are tallying with the real controls.

However, construction professionals, due to a lack of knowledge of the controls, did not comment on the features. Even though the excavator operators agreed on the accuracy of the controls, 65% ($f = 9$) of them denied the option of considering EAR as a basic platform to learn excavator controls as suggested by the construction professionals (67% or $f = 10$). A chief engineer of a large-scale construction project pointed out:

[. . .] from this app, we can obtain a basic understanding of an excavator control as you have included necessary controls. (R13)

Whereas an excavator operator experienced in the field stated:

[. . .] hands-on experience is compulsory. We need to learn from experience because, in reality, the process is much tougher [. . .] deep excavations, deep slopes are some examples. (R20)

A similar view was provided by trainers (R15, R16) who insisted on the need for hands-on experience. However, R16 pointed out that improvements made to the EAR to reflect real site challenges and experience would make it a platform for training experiences.

4.3.2 Weaknesses. 84% ($f = 27$) of the respondents acknowledge that the EAR App should incorporate more features to reflect many aspects, such as safety and auditory

aspects. Surprisingly more than half of the respondents representing each category agreed with such a requirement. Ignoring site complexities represented the second-highest referred weakness, among which construction professionals displayed the highest level of agreement (87% or $f = 13$) compared to the other participants.

R1 pointed out:

[...] despite the soil tests [...] there was a major difference in soil texture and content [...] Once an underground sewer pipeline was damaged during an excavation process.

However, ignoring changes in the environment ranked third, with 59% of the agreement pointed out by the majority of excavator operators (64% or $f = 9$) and trainers compared to construction professionals (53% or $f = 08$). Excavator operators and trainers have hands-on experiences indicating similar opinions.

This EAR is an original application developed to improve construction labours' and excavator operators' skills and help them identify hazards in various construction sites. EAR is an online application that can be installed on tablets and smartphones for individuals. However, the EAR is limited to an individual per experimentation and does not allow multiple users or participants to present in the same experience. This can be considered as a future task of upgrading the EAR.

4.3.3 Opportunities for improvement. By considering the current strengths and weaknesses of the EAR, respondents suggested areas for improvement in various domains. Incorporating emergency and safety features were reflected among 91% of the respondents securing the majority in each respondent category. Examples include:

[...] in case of an emergency, how to operate? as an example, if there is a massive drop, then how to operate? (R5, Equipment manager).

[...] in actual site situations if we can visualise the places of accidents for example, when working with the excavator, where we should place the barrier? where should the watchmen be positioned to monitor? what possible incidents can happen when moving the buckets? How people in the vicinity should be considered? (R9, Health and Safety Officer).

The need to include a self-levelling system to obtain a similar level of consensus (91% or $f = 29$) with more than 2/3 agreement among each respondent category. Examples of opinion include "change of level or slopes should be reflected in EAR" (R31). The need to consider space limitation received considerable attention among respondents, with the majority agreement between excavator operators and trainers in comparison to construction professionals.

The requirement for the EAR App to visualise the distance to obstacles was highlighted by the majority of construction professionals (93% or $f = 14$). Excavator operators and trainers agreed with such requirements. 78% of the construction professionals stated the inclusion of excavator categories, such as backhoe excavators and crawler excavators, would enhance the usage of the EAR app. However, only a few, limited to 21% ($f = 03$) of operators and no trainers, had a similar view. Their contradictory view included "it would be the app more complex" (R15) and "should have a step-by-step process to support training" (R26). A typical excavator is being used in different contexts in addition to excavation. These include material handling and forestry work. Therefore, 47% ($f = 15$) of the respondents stated the necessity of including these multiple tasks in the EAR app. Moreover, converting to a gaming app and synchronising a ground scanner were identified as other opportunities for improvement.

4.4 Challenges in application of Excavator Augmented Reality mobile app: political, economic, sociocultural, technological, environmental and legal analysis

During the study, it was evident that AR-based mobile applications are relatively new to practitioners in the Sri Lankan context and that a considerable gap exists in adoption. Therefore, to understand the threats prevailing in adopting AR-based applications, a PESTEL analysis was conducted.

4.4.1 Political challenges. Political tension between parties was highlighted by many respondents as a possible challenge from the dimension. Accordingly, R² stated: “if the introduction of technological platforms influences the parties with political support, there will be major challenges”. Moreover, several other respondents suggested that the hesitation to adopt (triggered by high initial cost and unclear benefits) might discourage construction companies from implementing or investing in the development of apps. Examples of a few quotes include “we need management consent” (R²), “there might be delays in approval” (R14); and “[. . .] we might say that there are other requirements for safety” (R12). As per the respondent’s views, the top management (directors and CEO), including the construction and safety manager’s commitment to digitalising construction, is necessary. Therefore, the study manifested that to solve the challenges arising from political spheres, the government should play a crucial role. This is through enforcing a clear vision of technological adoption and encouraging investors to follow the road map. This might motivate the top management towards a digitalised approach considering the investors’ requirements.

4.4.2 Economic challenges. One of the main highlighted economic challenges by the respondents is the cost associated with the application. An example includes the statement made by respondent R12, namely: “this app should be further developed and brought into a more comprehensive form first and then should be integrated with excavator units which might be very costly”. Further, respondents acknowledged the necessity of organisational and process change required to integrate a comprehensive app into the excavation process, namely: “[. . .] the cost of change in practices” (R8), “time and effort taken to adopt these change [. . .]” (R17) and “would the change be worthwhile” (R7). To overcome the challenges, the existence of sufficient funds and implementing a change management process by obtaining stakeholders’ buy-in were considered crucial by respondents.

4.4.3 Social challenges. Low educational background, resistance to change and the operator’s true potential getting hidden were pointed out as the main social challenges arising. According to R13:

[. . .] operators only have minimum educational level such as O/L [. . .] If we show something like this, they will think it is IT related and they become nervous. So their real capacity might get hidden.

Moreover, a lack of awareness of the technological advancements was considered another challenge, which would impact the successful implementation of AR apps. To overcome social challenges, knowledge management was considered a key tool by many construction professionals. Further, spreading awareness and encouraging self-education through participating in events such as continuing professional development and webinars were pointed out by construction professionals and trainers. Moreover, excavator operators suggested that training programs should be advanced and technological education should be improved to support the introduction of such apps.

4.4.4 Technology challenges. Lack of technical support, including reference to design architecture, was considered a foremost challenge in hindering AR adoption. Moreover, some respondents, such as R11, R14, R25 and R32 pointed out that the lack of the latest equipment and the existence of machinery in degraded condition would be another

challenge. “Our machinery is tool old” (R19) and “this machine was purchased 22 years back” (R21) are a few cases of opinions made by respondents. Further, R6, in particular, questioned the possibility of security breaches overriding the controls:

In the current form, there won't be any problem, but we will need to improve it for practical implementation like [...] in those instances what will happen if a security breach takes place.

To overcome these challenges, respondents suggested that a background enabling such development should be created.

4.4.5 Environmental challenges. Most of the respondents manifested the utilisation of AR apps would not create environmental challenges such as pollution. However, safety officers noted the necessity of looking into legal compliance. Moreover, they commented in relation to the current challenge in terms of the legal background, which is being discussed under legal challenges.

4.4.6 Legal challenges. Lack of laws was agreed by the majority of the respondents as a key challenge of incorporating such apps in Sri Lanka. R10 commented:

[...] no law relating to construction sites, only we use factory ordinance. Even for scaffolding, we use British standards [...] Even without those basic requirements, I don't believe there will be legal laws enforced to support digitalises techniques like this.

Therefore, it is proved that a proper legal background in terms of technology adoption should be created for the introduction and sustainable implementation of AR-assisted apps in the construction industry. These issues were recently discussed by [Edwards et al. \(2021\)](#), who noted the difficulty in attributing liability in the event of an accident, i.e. does blame rest with the operator, the programmer, the software developer of the app user? It was concluded that existing legal frameworks must keep abreast with the rapid pace of technological development.

5. Discussions

AR has received a considerable amount of attention within the research and construction industry in the past two decades. However, the application of MAR in the excavation process has received comparatively scant attention. To fill this gap, the current study focuses on assessing such an application through a usability study of a novel MAR application called EAR for the excavation process in construction. Hence, the initial question addressed by the study was:

RQ1. What are the expectations and current level of satisfaction in relation to the developed EAR?

The study's findings revealed the highest satisfaction gap (refer to subsection 3.4) was demonstrated for the auditory aspect, where the respondents' expectations were far beyond their satisfaction levels. This aspect was followed by the usefulness of the App in learning construction equipment capability and received a 0.9 satisfaction level. The satisfaction gap for the visual aspects was ranked third with 0.8125 and the 4th, 5th, 6th, 7th, 8th, 9th, 10th and 11th ranks did not demonstrate a major satisfaction gap.

The second question posed was:

RQ2. What are the implications of EAR App to the construction industry?

The study's findings revealed that the EAR app could contribute to construction through three domains, namely: productivity, training and safety. Under training, the study

identified the app in its current form would: facilitate obtaining an initial exposure to excavator operation and controls; provide a platform to visually demonstrate the task; and act as a competency tool for supervisors to mentor and assess an operators' capability and familiarity with controls. On a similar note, [Lovreglio and Kinatader \(2020\)](#) stated that AR could provide effective, flexible and affordable training platforms for safety-relevant scenarios. Similarly, [Li *et al.* \(2018\)](#) stated that AR technologies have opened up new possibilities for effective training and education with a higher level of understanding and fewer risks. Notably, these past studies have not extensively investigated the usage of MAR for the excavation process and its implications.

The third research question addressed by the study was:

RQ3. What are the strengths, weaknesses and opportunities for improvement of the EAR app?

Examining the findings in relation to the strength, weaknesses and opportunities of the EAR app, several opportunities were predominant compared to strengths and weaknesses. The main strengths explored are visualising the excavator activities, accurate controls and a basic platform to learn controls. The overall participation and motivation of the respondents stand evidence of the overall effectiveness of the EAR app in expediting a meaningful learning environment. According to [Omar *et al.* \(2019\)](#), one of MAR's major advantages is its inherent ability to enhance visualisation skills, which is a requirement to succeed in a STEM field. The basic advantages of MAR, namely, visualisation and positive impact on learning, are apparent in the EAR app, suggesting the success of the application.

The weakness associated with the EAR app was next investigated and several features identified have been overlooked, including site complexities and environment changes. Though there are no specific studies that have developed MAR for the excavation process, several studies carried out by [Nincarean *et al.* \(2013\)](#), [Kounavis *et al.* \(2012\)](#) and [Choudary *et al.* \(2009\)](#) in the context of education, tourism, culture, respectively had provided high prominence for external environmental factors and improved features. Hence, the weaknesses of the EAR app should be overcome through further studies in the area. To facilitate the purpose, opportunities for improvements were examined by the present study – for example, including emergency and safety features (self-levelling system, space limitations and obstacle detection), the inclusion of excavator categories; supporting multiple purposes; converting to a gaming app and synchronise a ground scanner.

The final research question was:

RQ4. What are the political, economic, social, technological, environmental and legal challenges associated with such applications and solutions?

To address this question, the study carried out a PESTEL analysis. Even though as a whole PESTEL analysis has not been carried out relevant to MAR for excavation, several studies exist covering individual parameters. [Ramilo and Embi \(2014\)](#) and [Chowdhury *et al.* \(2019\)](#) stated in their study on the adoption of new technologies that government barriers hinder the overall acceptance of new technologies. Similarly, in a survey carried out in Singapore ([Ramilo and Embi, 2014](#) and [Ozorhon *et al.* \(2010\)](#) in their study derived that the most significant barrier to technology adoption is financial concerns. Therefore, it is evident that financial hindrance has become the main challenge in the overall progression of advanced technology adoption throughout the industry.

Low educational background, resistance to change and operator's true potential getting hidden and lack of awareness of the technological advancements were pointed out by

respondents as social challenges. Correspondingly, insufficient technical capabilities and skill levels, reluctance to change, inexperienced team members and lack of training were mainly identified as social challenges by authors such as [Cory and Bozell \(2001\)](#) and [Jones and Saad \(2003\)](#) in technological adoption. This justifies those social challenges that are not unique to MAR implantation but have become a burden towards the entire introduction of technologies to the construction sector.

Technical challenges to EAR app adoption included: lack of technical support, lack of the latest equipment and/or existing machinery being in a degraded condition. [Ramilo and Embi, \(2014\)](#) found out that inadequate R&D investment, inadequate technology transfer and inflexible building codes and standards as the main technological challenges. Hence, it is evident that the overall traditional technological challenges hindering the construction industry prevail even in implementing the latest technologies such as AR, suggesting the backward nature of the industry.

In terms of environmental challenges, the study derived that the introduction of the EAR app would not possess any environmental challenges. Similarly, many studies that investigated technological adoption for construction had no concerns about challenges derived from the environmental sphere proving the findings of the study ([Delgado et al., 2019](#)). It could be concluded that the outcome of the PESTEL analysis carried out for the EAR app derives several similar challenges compared to the challenges in technological adoption in the construction sector.

6. Conclusions

This paper contributes to the body of knowledge by identifying key measures of human-machine interaction in an immersive space using a novel EAR. The novelty of the EAR is its online features which are freely accessible to users, including participants in developing countries, and it can be easily installed on smartphones and tablets. This paper also addressed a significant gap in the literature on how AR mobile apps can enhance safety where the job-site task involves the interaction of humans and machinery. This study presented the outcome of a series of experiments to measure the usability of an AR-based mobile app for the excavation process called EAR in the Sri Lankan construction sector. In an attempt to understand the current user expectations and level of satisfaction with regard to the developed EAR app, it was found that the highest satisfaction gap exists for the auditory aspects. Among the remaining criterion, the satisfaction met user expectations for comfortability and no-risk practice. A satisfaction gap was evident among other criteria yet; the satisfaction gaps between means were below 1. Moreover, the study's findings derived that the main three implications of the EAR app are focused on enhancing productivity, training and safety in the construction sector. However, the existence of a satisfaction gap implied the need for developments in the EAR app, which was examined through SWOT analysis. To examine the threats to implementing the EAR app, the study carried out a PESTEL analysis and revealed that several challenges prevail from political, economic, social, technological and legal perspectives.

With the continuous integration of AR technology into the construction industry, the current study would set a path for construction professionals to understand the user expectation of MAR apps. Software developers can consider important factors identified through this study, such as ease of use and the perceived enjoyment of the software, to create MAR platforms that have a higher possibility to be used and accepted by the users. The EAR application was limited to individual experimentation and future studies can extend the application to be multi-user and a group of participants can be present all together in the experimentation and communicate their ideas while using the application. No

studies have been carried out so far to assess a MAR application such as EAR for the excavation process. Hence, the scope of the study is limited to excavation, given the emerging requirement and the existing gap. The study is limited to testing the usability of the EAR app within the construction sector of Sri Lanka. Hence, further studies can be carried out in validating the current study results in other countries; testing the applicability of the findings to other construction-related activities; and refinement of the EAR app based on the user requirements and expectations.

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Corresponding author

Fathima Nishara Abdeen can be contacted at: n.abdeen@unsw.edu.au